

Shared-Line Multitone SQR in Dispersive cQED: A Controlled No-Go, the Decoupled-Block Exception, and Why Echo Does Not Rescue the Gate

Autonomous cQED Research Platform
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This study examines a deliberately narrow question: whether an ideal simultaneous selective qubit rotation can be realized exactly in dispersive circuit quantum electrodynamics when a shared qubit-control line carries one resonant tone per addressed cavity manifold, no artificial per-tone detuning is allowed, and the only free corrections are per-tone amplitude and azimuth. Several earlier local studies were suggestive but did not answer this exact question because they allowed frequency offsets or moved to richer waveform families. In the controlled block-resolved dispersive limit, a Magnus analysis shows that spectator tones generate blockwise effective Z terms. For the two-block square-pulse problem the second-order coefficients are explicit, and forcing both blocks to be pure XY rotations requires nongeneric extra relations beyond the amplitude and azimuth controls needed to match the desired transverse rotations. Exact shared-line simulations using the local `cqed_sim` framework then attempt to falsify that no-go claim. Across forty-four strict cases covering two model variants, addressed windows of size 2 to 4, three durations, and aligned, structured, and random target families, the best restricted average gate fidelity reaches only 0.8058, with a mean of 0.6094. An exact reduced blockwise replay of the same shared-line waveform agrees with the full propagation to machine precision, showing that the failure is already present in the block-resolved problem rather than being caused by leakage. By contrast, a stronger decoupled-block approximation that drops all spectator tones exactly realizes the ideal target with unit fidelity by construction, but it is no longer the same simultaneous shared-line control problem. Finally, the echoed sequence half-SQR $\rightarrow \pi \rightarrow$ half-SQR $\rightarrow \pi$ suppresses some residual Z accumulation only at first order. Ideal instantaneous echoes reduce the mean matched-set residual- Z error from 7.86×10^{-2} rad to 1.35×10^{-2} rad, but the mean fidelity still falls from 0.7133 for the plain pulse to 0.2018, and finite 40 ns Gaussian π pulses perform worse still. The main conclusion is therefore negative but precise: exact ideal simultaneous shared-line multitone SQR is generically unavailable under the strict no-detuning amplitude-plus-azimuth ansatz, while exact realizability reappears only after strengthening the approximation enough to change the physical problem.

I. INTRODUCTION

Selective qubit rotations conditioned on cavity Fock number are a standard control primitive in dispersive circuit quantum electrodynamics (cQED) [1–3]. The question studied here is narrower than the usual gate-synthesis problem. We ask whether an *ideal* simultaneous multitone selective qubit rotation can be produced exactly when one drives a shared qubit-control line with one tone per addressed manifold, pins every tone to the corresponding dispersive transition frequency, forbids any artificial per-tone detuning, and allows only per-tone amplitude and azimuth corrections.

That question matters because nearby studies in this repository already explored closely related multitone constructions, including direct selective rotations, echoed sequences, and richer parameterized waveforms. Those studies are useful prior art, but after auditing them carefully we found that they do not settle the strict question posed here. The closest positive-looking cases either optimized per-tone frequency offsets, thereby introducing an explicit compensating Z knob, or studied broader waveform families that are outside the strict simultaneous shared-line ansatz. A skeptical re-examination is therefore needed before any exact-realizability claim is trusted.

The present report separates four logically distinct statements.

1. In the controlled dispersive block-resolved model, simultaneous shared-line multitone driving generates unwanted blockwise Z terms at second Magnus order.
2. Under the strict amplitude-plus-azimuth control ansatz, canceling those Z terms while also matching arbitrary target transverse rotations is generically impossible.
3. A stronger decoupled-block approximation *does* permit ideal blockwise selective rotations, but that approximation is physically different because it removes the shared-line spectator problem by assumption.
4. A multitone echo sequence can suppress some Z accumulation only approximately and under narrower assumptions than an exact ideal-gate claim would require.

II. MODEL, TARGET, AND SCOPE

We use the block-resolved dispersive Hamiltonian

$$H_0 = \sum_n \frac{\omega_n}{2} \sigma_z \otimes |n\rangle\langle n|, \quad (1)$$

with manifold frequencies

$$\omega_n = \omega_q + \chi n + \chi' n(n-1) + \dots \quad (2)$$

The control drive is a shared qubit-line multitone pulse

$$H_d(t) = \sum_{m \in \mathcal{S}} \frac{\Omega_m f(t)}{2} \left(e^{-i(\omega_m t - \phi_m)} \sigma_+ + e^{i(\omega_m t - \phi_m)} \sigma_- \right), \quad (3)$$

where $\mathcal{S}$ is the addressed Fock set, ω_m is chosen *exactly* at the corresponding manifold transition, $f(t)$ is the common envelope, and there is no independent per-tone detuning or added Z -compensation term. In the main strict study we take $f(t) = 1$ on $0 \leq t \leq T$, so the pulse is a simultaneous square multitone burst.

The target gate is an ideal Fock-block-diagonal selective rotation

$$U_{\text{SQR}}^{\text{ideal}} = \bigoplus_{n \in \mathcal{S}} R_{\hat{n}(\varphi_n)}(\theta_n), \quad (4)$$

with

$$R_{\hat{n}(\varphi)}(\theta) = \exp \left[-\frac{i\theta}{2} (\cos \varphi X + \sin \varphi Y) \right]. \quad (5)$$

Every addressed block must therefore end as a pure XY -plane rotation with no residual block-dependent Z phase.

The analytical baseline deliberately assumes the dispersive block structure is exact and neglects inter-Fock leakage. That assumption is not hidden: later numerical sections check it directly by comparing the full shared-line propagation against an exact reduced blockwise replay of the same waveform. The role of higher-order dispersive curvature is also isolated explicitly by running both a χ -only model and a $\chi + \chi'$ model.

III. CONTROLLED NO-GO FOR THE SIMULTANEOUS SHARED-LINE PROBLEM

A. Blockwise interaction-frame Hamiltonian

Let $\lambda_m = \Omega_m/2$, $\Delta_{nm} = \omega_m - \omega_n$, and

$$X_\psi = \cos \psi X + \sin \psi Y. \quad (6)$$

In the interaction frame of H_0 , block n sees

$$H_I^{(n)}(t) = f(t) \sum_{m \in \mathcal{S}} \lambda_m X_{\phi_m - \Delta_{nm} t}. \quad (7)$$

The resonant term $m = n$ is static and transverse. Every spectator tone $m \neq n$ remains in the same block, but

rotates in the XY plane at beat note Δ_{nm} . The key point is that shared-line simultaneity keeps all of those terms present at once.

For a square pulse, the first Magnus average is

$$\bar{H}_n^{(1)} = \lambda_n X_{\phi_n} + \sum_{m \neq n} \lambda_m \operatorname{sinc} \left(\frac{\Delta_{nm} T}{2} \right) X_{\phi_m - \Delta_{nm} T/2}, \quad (8)$$

with $\operatorname{sinc}(x) = \sin(x)/x$. At this order the spectators only distort the intended transverse axis and angle. A residual Z term first appears at second Magnus order.

It is convenient to write

$$\beta_n(t) = f(t) \sum_{m \in \mathcal{S}} \lambda_m e^{-i(\Delta_{nm} t - \phi_m)}, \quad (9)$$

so that

$$H_I^{(n)}(t) = \beta_n(t) \sigma_+ + \beta_n^*(t) \sigma_-. \quad (10)$$

The exact second-order average-Hamiltonian contribution in block n is then

$$\bar{H}_n^{(2)} = \zeta_n Z, \quad \zeta_n = \frac{1}{T} \Im \int_0^T dt_1 \int_0^{t_1} dt_2 \beta_n(t_1) \beta_n^*(t_2). \quad (11)$$

This is the obstruction term: unless $\zeta_n = 0$, block n is not a pure XY rotation.

B. Explicit two-block proof

The cleanest derivation is the two-block case $\mathcal{S} = \{0, 1\}$ with $\Delta = \omega_1 - \omega_0 > 0$ and $\delta = \phi_0 - \phi_1$. Evaluating Eq. (11) for a square pulse gives

$$\zeta_0 = -\lambda_1^2 K(\Delta, T) + \lambda_0 \lambda_1 L(\Delta, T, \delta), \quad (12)$$

$$\zeta_1 = +\lambda_0^2 K(\Delta, T) - \lambda_0 \lambda_1 L(\Delta, T, \delta), \quad (13)$$

where

$$K(\Delta, T) = \frac{\Delta T - \sin(\Delta T)}{\Delta^2 T}, \quad (14)$$

and

$$L(\Delta, T, \delta) = \frac{\Delta T [\cos \delta + \cos(\Delta T + \delta)]}{\Delta^2 T} + \frac{2 \sin \delta - 2 \sin(\Delta T + \delta)}{\Delta^2 T}. \quad (15)$$

The function $K(\Delta, T)$ is strictly positive for $\Delta > 0$ and $T > 0$, because $x - \sin x > 0$ for every $x > 0$. It is therefore not removable by a phase choice. Equations (13) also show the full second-order structure clearly:

- the $\lambda_1^2 K$ and $\lambda_0^2 K$ terms are spectator self-terms and do not depend on azimuth,
- the L term is an interference correction that does depend on the phase difference.

Now impose the ideal-gate requirement $\zeta_0 = \zeta_1 = 0$ with both addressed rotations nontrivial, so $\lambda_0\lambda_1 \neq 0$. Equations (13) imply

$$\lambda_0 L = \lambda_1 K, \quad \lambda_0 K = \lambda_1 L. \quad (16)$$

Multiplying and subtracting gives

$$\lambda_0^2 = \lambda_1^2, \quad L(\Delta, T, \delta) = K(\Delta, T). \quad (17)$$

That is the clean two-block obstruction. In the weak-drive dispersive regime, the first Magnus term fixes the desired transverse angles as $\theta_n = 2\lambda_n T + O(\lambda/\Delta)$. For a generic target with $\theta_0 \neq \theta_1$, equal amplitudes are already incompatible with matching both rotations. Even in the special equal-angle case, one must still land on the codimension-one relation $L = K$. The cancellation set therefore has no open interior in the space of nontrivial two-block targets and durations.

C. Many-block generalization and the meaning of “generic”

For N addressed blocks, Eq. (11) yields N analytic constraints

$$\zeta_n(\{\lambda_m, \phi_m\}, T) = 0, \quad n \in \mathcal{S}, \quad (18)$$

in addition to the $2N$ conditions needed to match the desired blockwise XY angles and azimuths. In the controlled dispersive regime the first-order map from (λ_n, ϕ_n) to (θ_n, φ_n) is locally invertible away from trivial zero-angle points, so the transverse target already fixes the $2N$ available controls locally. The equations $\zeta_n = 0$ therefore add N further real conditions without providing new control knobs.

This is what “generically impossible” means in the present report: exact ideal simultaneous shared-line multitone SQR is not available on any open set of nontrivial target specifications and durations. Exact cancellation can still occur on accidental lower-dimensional subsets, for example when special equal-angle or phase-duration relations hold, but those sets are fine tuned rather than structurally stable.

The conclusion is deliberately phrased as a controlled no-go, not as an all-regime nonperturbative theorem. The derivation uses the block-resolved dispersive model and a Magnus expansion [4]. The next section therefore attempts to falsify the no-go claim with exact shared-line propagation rather than assuming the perturbative argument is automatically sufficient.

IV. NUMERICAL ATTEMPTED FALSIFICATION

A. Simulation design

All strict simulations use the local `cqed_sim` framework [5]. The multitone waveform is propagated

TABLE I. Headline comparison of the main constructions. The fidelity is the restricted average gate fidelity on the addressed block subspace.

Construction	Mean	Best	Mean max Z [rad]
Full shared-line	0.6094	0.8058	0.1278
Exact reduced	0.6094	0.8058	0.1278
Magnus	0.2310	0.4405	1.2665
Decoupled	1.0000	1.0000	0
Echo, ideal	0.2018	0.2077	0.0135
Echo, finite	0.3366	0.4418	0.4146

through the exact shared-line dispersive model rather than through the truncated Magnus approximation. The optimizer is intentionally restricted to amplitude and azimuth corrections only; per-tone detuning remains fixed at zero throughout.

The numerical sweep covers two Hamiltonian variants, two deterministic target families and one random family, addressed windows of size $N_{\text{active}} = 2, 3, 4$, and three durations,

$$\frac{|\chi|T}{2\pi} \in \{1, 3, 5\}, \quad (19)$$

which correspond to $T \approx 352$ ns, 1056 ns, and 1761 ns for $|\chi|/2\pi = 2.84$ MHz. The default Hilbert-space truncation is two transmon levels and $N_{\text{active}} + 2$ cavity levels. A representative convergence case is then rerun with smaller timestep, larger optimization budget, and one extra transmon level.

The study does not rely on a single scalar fidelity. For each case we record the restricted process fidelity, restricted average gate fidelity, per-block axis-angle reconstruction, per-block residual Z error, per-block transverse error, and population outside the targeted Fock blocks. We also compare the full shared-line result against an exact reduced blockwise replay that keeps the same compiled waveform but solves each Fock block separately. That comparison is crucial because it tests whether the no-go is already present in the block-resolved physics or is instead an artifact of leakage.

B. Headline results

Table I summarizes the main constructions. The strict simultaneous shared-line problem fails broadly: the mean restricted average gate fidelity is only 0.6094, the best case reaches 0.8058, and the worst falls to 0.3011. The exact reduced blockwise replay matches the full result to machine precision, with mean full-versus-reduced fidelity 1.000000. The failure is therefore not caused by population leaving the addressed blocks; it is already present in the dispersive block-resolved shared-line dynamics.

Figure 1 shows the duration dependence for the two deterministic target families. The decoupled-block approximation remains exact by construction, while the strict

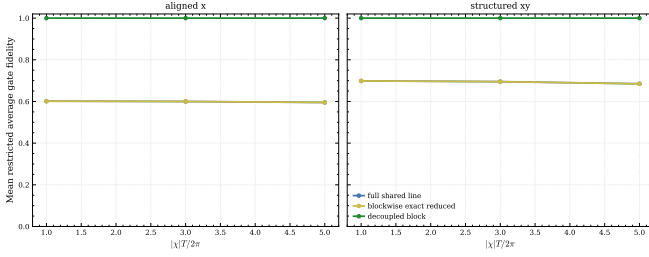


FIG. 1. Mean restricted average gate fidelity versus normalized duration. The strict simultaneous shared-line problem never reaches the ideal target, while the stronger decoupled-block approximation remains exact. The exact reduced blockwise replay sits directly on top of the full shared-line curve, confirming that the failure is already present in the block-resolved model.

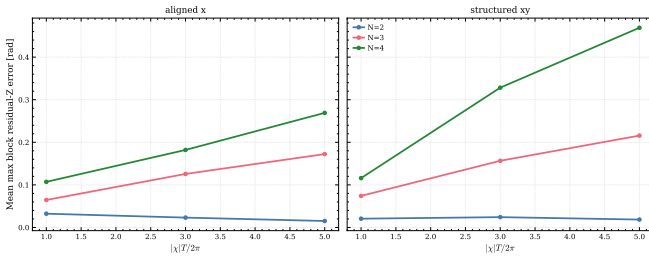


FIG. 2. Mean maximum blockwise residual- Z error versus normalized duration for the strict shared-line model. The residual phase grows with both duration and the size of the addressed manifold window.

shared-line fidelity decreases as the pulse grows longer. Averaged over all strict shared-line cases, the mean fidelity decreases from 0.6500 at $|\chi|T/2\pi = 1$ to 0.5689 at $|\chi|T/2\pi = 5$. The mean maximum residual- Z error also grows with duration, from 6.91×10^{-2} rad to 1.70×10^{-1} rad.

Figure 2 shows how the residual- Z error increases with addressed-window size and duration. The mean strict shared-line fidelity falls from 0.6999 at $N_{\text{active}} = 2$ to 0.5109 at $N_{\text{active}} = 4$. The same trend appears in the mean maximum blockwise residual- Z error, which grows from 2.20×10^{-2} rad to 2.45×10^{-1} rad over the same window-size sweep.

The blockwise diagnostics also show that residual Z is not the whole practical error budget. Across the strict shared-line set, the mean best-fit block-gauge improvement in process fidelity is only 2.30×10^{-3} , whereas the mean transverse axis-angle mismatch is 1.43 rad. In other words, once the optimizer is forced to use only amplitudes and azimuths, it cannot simultaneously remove the spectator-induced Z tendency and preserve the intended transverse action.

C. Why the numerical study supports, rather than replaces, the no-go

The numerical section is framed as an attempted falsification. If the analytical obstruction were merely a poor perturbative artifact, one would expect the exact shared-line optimization to find near-unit-fidelity counterexamples. That does not happen. Instead, every tested case remains far below unit fidelity, the exact reduced replay reproduces the same failure, and adding χ' only changes the strict shared-line mean fidelity from 0.6147 to 0.6041. The basic obstruction is therefore already visible in the minimal dispersive model.

The second-order Magnus approximation is intentionally kept in the data products as a qualitative reference, but it is not numerically accurate enough to stand alone at the tested drive strengths: its mean fidelity to the full shared-line result is only 0.3350. That is why the report does not claim more than the analysis can support. The Magnus expansion identifies the obstruction; the exact shared-line propagation shows that the obstruction survives beyond the perturbative surrogate.

V. THE STRONGER DECOUPLED-BLOCK APPROXIMATION

Now impose the stronger approximation requested in the prompt: in block n , keep only the resonant tone for that block and drop all off-resonant tones exactly. The block Hamiltonian becomes

$$H_{n,\text{dec}}(t) = \lambda_n f(t) X_{\phi_n}. \quad (20)$$

For a square pulse of duration T ,

$$U_{n,\text{dec}} = \exp[-iT\lambda_n X_{\phi_n}] = R_{\hat{n}(\phi_n)}(2\lambda_n T). \quad (21)$$

Therefore the ideal target is realized exactly by choosing

$$\lambda_n = \frac{\theta_n}{2T}, \quad \phi_n = \varphi_n. \quad (22)$$

The strict no-go does not apply because the spectator terms that generated Eq. (11) have been deleted by assumption.

This success must not be misinterpreted. The decoupled-block model is no longer the same simultaneous shared-line multitone problem. Each addressed block now evolves under a different effective Hamiltonian that never sees the other tones. Operationally, that corresponds either to addressing the manifolds one at a time or to assuming perfect decoupling so strong that the shared-line spectator problem has disappeared. It is a valid and sometimes useful approximation, but it is a stronger approximation, not a proof that the physical simultaneous problem was always solvable.

The numerics make that distinction explicit. The decoupled-block construction reproduces the ideal target with unit fidelity in every case, while the full shared-line

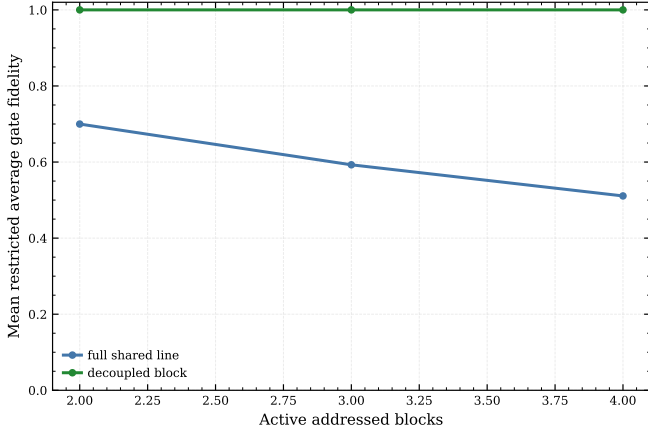


FIG. 3. Mean restricted average gate fidelity versus the number of addressed blocks. The stronger decoupled-block approximation remains exact, whereas the simultaneous shared-line fidelity degrades as more blocks are addressed.

model falls further from unity as the addressed window grows. Figure 3 shows the contrast directly.

This section also explains why some earlier local studies could appear successful. If a calculation implicitly or explicitly removed spectator tones block by block, or if it added per-tone detuning to compensate the induced Z phase, it was no longer testing the strict shared-line no-detuning question studied here.

VI. ECHO ALTERNATIVE: WHAT IT CANCELS AND WHAT IT DOES NOT

A. Sequence convention and toggling-frame analysis

The proposed echo sequence is the natural shared-line analogue of a refocusing construction [6]:

$$U_{\text{echo}} = X_{\pi} U_{1/2} X_{\pi} U_{1/2}, \quad (23)$$

where $U_{1/2}$ is the same strict shared-line multitone ansatz optimized for the half-angle target and $X_{\pi} = R_x(\pi)$. We study two versions:

1. an idealized instantaneous- π model, and
2. a finite implementation with two 40 ns Gaussian qubit π pulses calibrated on the vacuum manifold.

The toggling algebra is immediate:

$$X_{\pi} X X_{\pi} = X, \quad X_{\pi} Y X_{\pi} = -Y, \quad X_{\pi} Z X_{\pi} = -Z. \quad (24)$$

Hence

$$X_{\pi} X_{\varphi} X_{\pi} = X_{-\varphi}. \quad (25)$$

If a half-segment were generated by $aX_{\varphi} + cZ$, the echoed second segment would therefore be generated by $aX_{-\varphi} - cZ$. Two consequences follow.

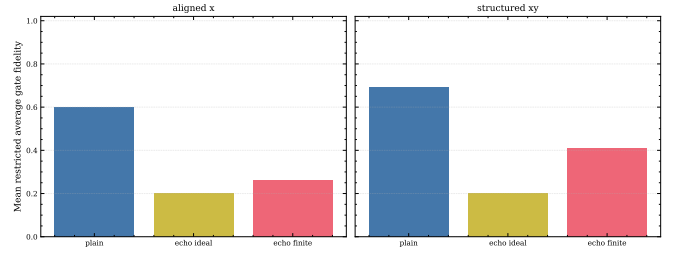


FIG. 4. Mean fidelity comparison for the plain strict multitone pulse and the two echoed variants. The ideal instantaneous echo reduces residual- Z accumulation but does not preserve the intended gate, and the finite Gaussian echo performs worse because the refocusing pulse is itself manifold dependent.

1. For a generic XY target with $\varphi \neq 0, \pi$, the desired transverse axis is not preserved exactly because the Y component flips.
2. Even for the aligned- x special case, the Z term is canceled only to first order. The two half-segment generators do not commute, so the exact combined propagator is not simply a full-angle pure X rotation.

This is the central reason the echo cannot be advertised as an exact rescue of the strict simultaneous multitone problem.

B. Numerical comparison

The numerical comparison bears out the toggling analysis. On the matched subset where echo was evaluated, the plain strict shared-line pulse has mean fidelity 0.7133. The ideal instantaneous echo suppresses the mean maximum residual- Z error from 7.86×10^{-2} rad to 1.35×10^{-2} rad, but the mean fidelity collapses to 0.2018. The transverse mismatch hardly changes, shifting only from 1.208 rad to 1.197 rad. The echo is therefore canceling the wrong piece of the problem: it reduces a part of the residual phase accumulation without repairing the overall gate action.

The finite Gaussian implementation is worse. Its two inserted π pulses add 80 ns to the total duration, which is a 4.5% to 22.7% overhead across the tested duration range, and they are themselves dispersively nonuniform once the cavity is populated. As a result, the mean maximum residual- Z error rises to 0.4146 rad, the mean transverse error rises to 1.964 rad, and the mean fidelity is only 0.3366. Not a single finite-echo case outperforms its plain strict counterpart.

Figure 4 summarizes the mean-fidelity comparison for the aligned and structured target families.

The echoed sequence therefore works only in a narrow and approximate sense: it can suppress some Z -type accumulation under idealized assumptions, but it does not

restore exact ideal shared-line multitone SQR, and realistic finite refocusing pulses make the situation worse rather than better in the present regime.

VII. FINAL VERDICT

The study prompt asked for four direct answers.

1. *Is exact ideal simultaneous shared-line multitone SQR possible under the physical no-detuning amplitude-plus-azimuth model?* Not generically. In the controlled dispersive block-resolved regime, the second-order Magnus calculation produces blockwise Z constraints that are not automatically removable once the transverse target is fixed. Exact shared-line simulations then fail to find any near-unit-fidelity counterexample: the best strict case reaches only 0.8058, and the mean is 0.6094.

2. *Under what assumptions is it impossible?* The no-go is derived under the shared-line dispersive block-resolved model with simultaneous tones, no artificial per-tone detuning, and the ideal target defined as a direct sum of pure XY rotations with no block-dependent Z phase. “Generically impossible” means the exact-cancellation set has no open interior in target-and-duration space; only fine-tuned lower-dimensional exceptions remain.

3. *Under what stronger approximation does ideal SQR become achievable?* If one strengthens the model so that each block retains only its own resonant tone and all spectator tones are dropped exactly, then each block becomes an independently driven qubit and the ideal target is realized by choosing $\lambda_n = \theta_n/(2T)$ and $\phi_n = \varphi_n$. That approximation is physically different from the simultaneous shared-line problem because it assumes perfect blockwise decoupling.

4. *Does the echoed multitone sequence rescue the situation exactly, approximately, or only in special regimes?* Only approximately, and only in a narrow sense. Ideal instantaneous echoing can suppress some Z -precession to first order, but it does not preserve the desired gate exactly; for generic XY targets the transverse axis is mirrored, and even the aligned- x case retains noncommuting error terms. Finite Gaussian refocusing pulses are worse still in the tested regime.

VIII. LIMITATIONS AND FUTURE WORK

The strongest analytical claim in this report is a controlled no-go inside the block-resolved dispersive model, not an all-regime theorem for arbitrary strong driving. The numerical study is extensive enough to be convincing within that model class, but it still samples a finite target family rather than every conceivable target. The optimizer is also deliberately restricted to the strict amplitude-plus-azimuth ansatz; that is a feature for the present question, not a statement that richer control families are uninteresting.

TABLE II. Selected aggregate trends for the strict shared-line model.

Aggregate slice	Mean fidelity	Mean max Z error (rad)
$N_{\text{active}} = 2$	0.6999	0.0220
$N_{\text{active}} = 3$	0.5928	0.1454
$N_{\text{active}} = 4$	0.5109	0.2452
$ \chi T/2\pi = 1$	0.6500	0.0691
$ \chi T/2\pi = 3$	0.6194	0.1295
$ \chi T/2\pi = 5$	0.5689	0.1700

TABLE III. Validation spot checks for the representative structured- XY three-block case at $|\chi|T/2\pi = 3$.

Check	Fidelity	Max Z [rad]
Default settings	0.696774	0.2636
Smaller timestep	0.697201	0.2355
Larger timestep	0.691103	0.3151
Three transmon levels	0.696958	0.2636

The next steps are therefore clear. First, if the physical goal is exact or near-exact realizability, one should study the minimum extra control resource needed to remove the obstruction cleanly, such as per-tone detuning, explicit Z -compensation, or genuinely time-segmented control rather than simultaneous shared-line square pulses. Second, if echoed constructions remain of interest, they should be reformulated with manifold-aware refocusing pulses or with a target family aligned to the actual toggling symmetry rather than to a generic XY axis. Third, the exact reduced blockwise replay and residual-generator diagnostics developed here would be worth upstreaming into the local simulation framework because they make future studies easier to audit.

Appendix A: Numerical Details, Validation, and Local Scope Audit

The strict shared-line sweep consists of 44 base cases: two Hamiltonian variants, two deterministic target families, one random target family, addressed windows $N_{\text{active}} = 2, 3, 4$, and durations $|\chi|T/2\pi = 1, 3, 5$. Echo was evaluated on the 24 deterministic cases with $N_{\text{active}} \leq 3$. The full production sweep required about 529 s on the local workstation.

The representative convergence case is the $\chi + \chi'$, structured- XY , three-block, $|\chi|T/2\pi = 3$ instance. Re-running that case with a larger optimization budget leaves the restricted average gate fidelity unchanged at 0.696774 to six decimal places. Changing the timestep from 2 ns to 1 ns shifts the restricted average gate fidelity only from 0.696774 to 0.697201, and increasing the transmon truncation from two to three levels changes it from 0.696774 to 0.696958. Those shifts are small compared with the gap to unit fidelity.

The local scope audit was equally important. The

nearest earlier studies in this repository did not answer the present question directly because they allowed per-tone frequency corrections, explored richer waveform families, or relaxed the notion of success away from an exact no-detuning ideal-gate test. Those studies are still useful context, but they should not be cited as evidence that the strict shared-line amplitude-plus-azimuth problem is exactly solvable.

Appendix B: Reproducibility

The machine-readable study outputs are stored under the study data and artifact directories. The main summary files are

- `study_results.json`,
- `study_results.csv`,

- `study_summary.json`,
- `analytic_summary.json`,
- `validation_summary.json`,
- `prior_audit.json`.

Per-case restricted operators, waveform samples, and echoed-sequence artifacts are stored in the artifact directory, and the required reproducibility notebook is provided in the scripts directory.

The shortest complete reproduction path is:

1. run the strict-study driver to regenerate the shared-line, reduced, decoupled, and echoed results,
2. run the validation script to regenerate the convergence and sanity summary,
3. rebuild the reproducibility notebook if needed,
4. compile this report from the report directory.

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| <p>[1] A. Blais, R.-S. Huang, A. Wallraff, S. M. Girvin, and R. J. Schoelkopf, <i>Physical Review A</i> 69, 062320 (2004).</p> <p>[2] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, <i>Reviews of Modern Physics</i> 93, 025005 (2021).</p> <p>[3] J. Koch, T. M. Yu, J. Gambetta, A. A. Houck, D. I. Schuster, J. Majer, A. Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf, <i>Physical Review A</i> 76, 042319 (2007).</p> | <p>[4] S. Blanes, F. Casas, J. A. Oteo, and J. Ros, <i>Physics Reports</i> 470, 151 (2009).</p> <p>[5] Shankar Quantum Circuits Group, cqed_sim: Circuit quantum electrodynamics simulation framework, https://github.com/SoraUmika/qubox_cQEDsim (2026), local framework checkout used for this study.</p> <p>[6] E. L. Hahn, <i>Physical Review</i> 80, 580 (1950).</p> |
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